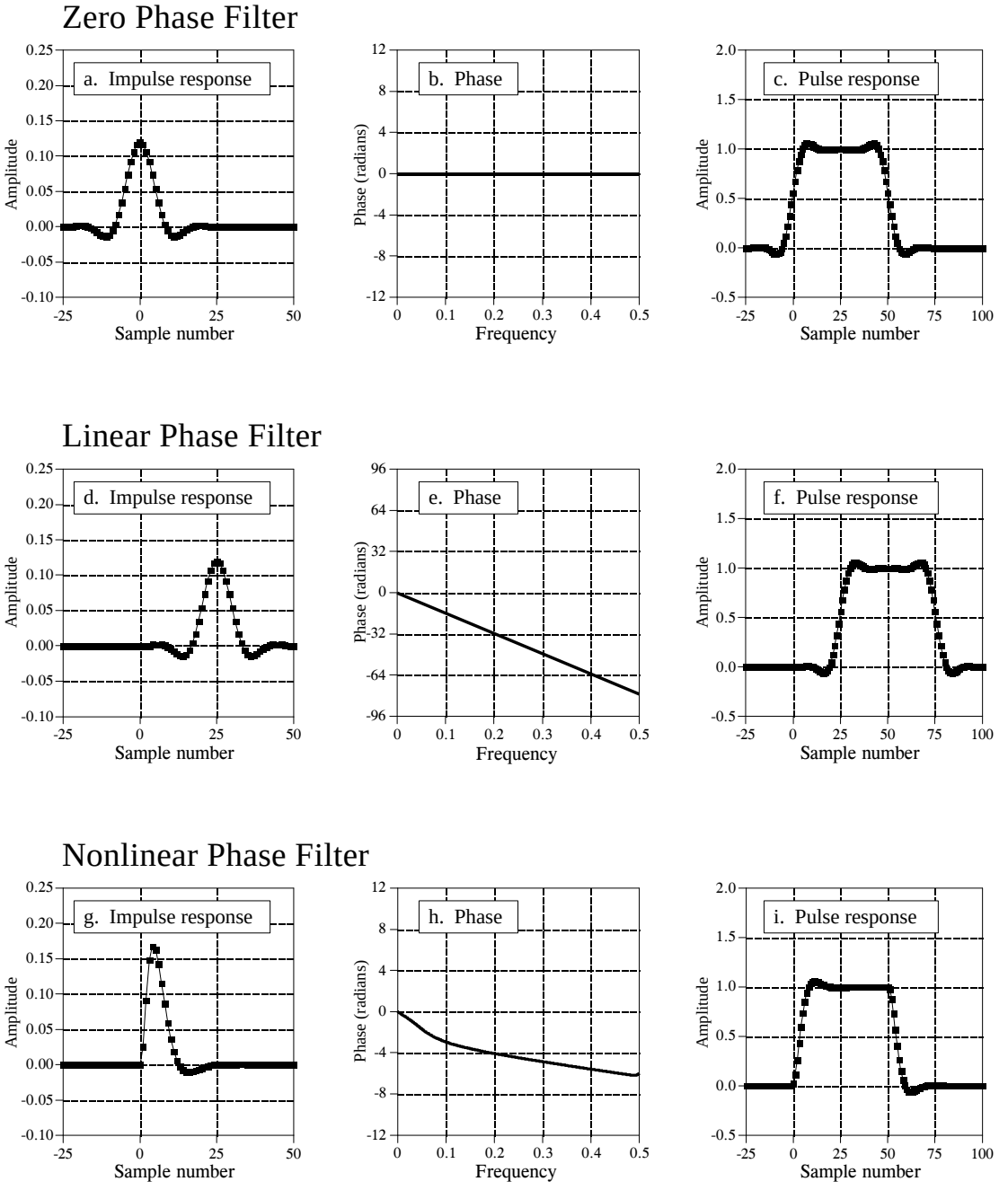

FIGURE 19-7
Zero, linear, and nonlinear phase filters. A *zero phase* filter has an impulse response that has left-right symmetry around sample number zero, as in (a). This results in a frequency response that has a phase composed entirely of zeros, as in (b). Zero phase impulse responses are desirable because their step responses are symmetrical between the top and bottom, making the left and right edges of pulses look the same, as is shown in (c). *Linear phase* filters have left-right symmetry, but not around sample zero, as illustrated in (d). This results in a phase that is linear, that is, a straight line, as shown in (e). The linear phase pulse response, shown in (f), has all the advantages of the zero phase pulse response. In comparison, the impulse responses of *nonlinear phase* filters are not symmetrical between the left and right, as in (g), and the phases are not a straight line, as in (h). The worst part is that the left and right edges of the pulse response are not the same, as shown in (i).